Tribhuvan University

Faculty of Management

OFFICE OF THE DEAN

2024

Bachelor in Business Administration

Course Title: **Business Statistics**

Code No: **STT201** Semester: **3rd Semester**

Full Marks: **60** Pass Marks: **24** Time: **s. The observed number of times for getting 4, 5 or 6 was recorded by the table below. Fit the binomial distribution for the given data.No. of face 4,5 or 601234Observed frequency210263515816.A committee of 5 is to be formed out of a group of 8 boys and 7 girls. Find the probability that in the committee.a. There will be all boys.b. There will be all girls.Section "C"Long Answer Questions: (Attempt any THREE Questions)[3 × 5 = 15]17.Calculate the percentile coefficient of kurtosis from the income distribution of sixty-two randomly selected employees of Nepal Bank Limited. Also interpret the result.Income (Rs '000)20-3030-4040-5050-6060-70TotalNo. of employees710201876218.The following table shows average marks of students in two classes.Class AClass BNo. of students150100Average marks4543Standard deviation of marks54Test whether there is any significant difference in average marks of students of two classes at 5% level of significance.19.The running capacity of two horses is given below, state which is more consistent and why?Horse A250255280290295300Horse B28028229029529829520.Students of a class are given a test. Their marks were normally distributed with mean 60 and standard deviation 5. What percentage of students scored:a. More than 65 marks.b. Between 50 and 65 marks.Section "D"Comprehensive Answer Questions:[20]21.The following table gives the marks in mathematics and statistics of ten students.Marks in mathematics45706530904050758560Marks in statistics35907040954060808050a. Calculate two regression coefficients.b. Calculate two regression lines.c. Find Karl Pearson's coefficient of correlation and test its significance.d. Estimate the most likely marks in statistics when the marks in mathematics is 52.**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

Brief Answer Questions:[10 × 1 = 10]

1. Find the quartile deviation if first and third quartiles are 18 and 30 respectively.

2. Calculate the combined mean from the following information.Group AGroup BMean6560Number of observations4060

3. The coefficient of correlation between two variable X and Y is 0.48. The covariance is 36. The variance of X is 16. Find the standard deviation of Y.

4. In a moderately asymmetrical distribution, the mode and mean are 32 and 35 respectively. Calculate the median.

5. Calculate the Pearson's Coefficient of skewness when mean, mode and standard deviation are 25, 20 and 10 respectively.

6. Given λ = 1.2 for a Poisson distribution, find p(x = 4).

7. If a random sample of size 36 is drawn from a finite population 400 units without replacement, then find the standard error of the sample mean if the population s.d. is 12.

8. Given that P(A) = 1/3, P(B) = 3/5 and P(A∩B) = 1/15 then find out the value of P(A ∪ B).

9. If first quartile and third quartile of a distribution are 28 and 52 and their 90th and 10th percentiles are 61 and 17 respectively then, find the value of kurtosis.

10. List out non-random sampling methods.

Section B

Short Answer Questions: (Attempt any FIVE Questions)[5 × 3 = 15]

11. The following table represents the weekly expenditure of 100 families.Expenditure0-2020-4040-6060-8080-100No. of Families14?27?15If the mean value is 48 find the missing frequencies.

12. A sample of 500 bulbs of a company shows that an average life of 1400 hours with standard deviation of 30 hours. Find 95% confidence limits for population mean.

13. Calculate coefficient of quartile deviation from the following distribution of expenditure of one thousand households of Pokhara city.Expenditure (Rs.'00)No. of households30-505450-7010070-9014090-110300110-130230130-150125Above 15051

14. The mark distribution of 104 students are given below.Marks0 - 1010 - 2020 - 3030 - 4040 - 5050 - 6060 - 70No. of Students7813293593Find the lowest marks of the top 10% of the students.

15. Five dice were thrown for 96 times. The observed number of times for getting 4, 5 or 6 was recorded by the table below. Fit the binomial distribution for the given data.No. of face 4,5 or 601234Observed frequency2102635158

16. A committee of 5 is to be formed out of a group of 8 boys and 7 girls. Find the probability that in the committee.a. There will be all boys.b. There will be all girls.

Section C

Long Answer Questions: (Attempt any THREE Questions)[3 × 5 = 15]

17. Calculate the percentile coefficient of kurtosis from the income distribution of sixty-two randomly selected employees of Nepal Bank Limited. Also interpret the result.Income (Rs '000)20-3030-4040-5050-6060-70TotalNo. of employees7102018762

18. The following table shows average marks of students in two classes.Class AClass BNo. of students150100Average marks4543Standard deviation of marks54Test whether there is any significant difference in average marks of students of two classes at 5% level of significance.

19. The running capacity of two horses is given below, state which is more consistent and why?Horse A250255280290295300Horse B280282290295298295

20. Students of a class are given a test. Their marks were normally distributed with mean 60 and standard deviation 5. What percentage of students scored:a. More than 65 marks.b. Between 50 and 65 marks.

Section D

Comprehensive Answer Questions:[20]

21. The following table gives the marks in mathematics and statistics of ten students.Marks in mathematics45706530904050758560Marks in statistics35907040954060808050a. Calculate two regression coefficients.b. Calculate two regression lines.c. Find Karl Pearson's coefficient of correlation and test its significance.d. Estimate the most likely marks in statistics when the marks in mathematics is 52.